2.Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

STEP:

Step 1: Download Datafile

Step 2: Load data file → pandas

Step 3: Display the Data File

Step 4: Check for missing values

Step 5: Fill or drop missing values

Step 6: Summary statistics

Step 7: Group by product and calculate the sales & quantity

Step 8: Bar plot of Total sales / product

Step 9: Line plot of sales over time

Step 10: Pivot table to analyze sales by region and product

Step 11: Correlation matrix

Step 12: heatmap of correlation matrix

PROGRAM:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

data = {

   "S.No": [1, 2, 3, 4, 5],

   "Date": ["2026-08-01","2026-08-02","2026-08-03","2026-08-04","2026-08-05"],

   "Product Name": ["Laptop","Mobile","Tablet",None,"Mobile"],

   "Sales": [55000,None,20000,60000,35000],

   "Quantity": [5,10,None,6,12],

   "Region": ["Chennai","Coimbatore","Madurai","Tiruchirappalli",None]

}

df = pd.DataFrame(data)

df.to_csv("sales_data.csv", index=False)

print("Data file created successfully!")
```

```
print("File name: sales_data.csv")

#LOAD DATA FILE

df = pd.read_csv("sales_data.csv")

print("\n DATA FILE LOADED ")

display(df)

#DISPLAY THE DATA FILE

print("\n SALES DATA ")

display(df)

#CHECK FOR MISSING VALUES

print("\nMISSING VALUES ")

missing_values = df.isnull().sum()

print(missing_values)

#FILL MISSING VALUES

df["Sales"] = df["Sales"].fillna(df["Sales"].mean())

df["Quantity"] = df["Quantity"].fillna(df["Quantity"].mean())

df["Product Name"] = df["Product Name"].fillna("Unknown")

df["Region"] = df["Region"].fillna("Unknown")

print("\n DATA AFTER HANDLING MISSING VALUES ")

display(df)

# SUMMARY STATISTICS

print("\nSUMMARY STATISTICS ")

display(df.describe())

#GROUP BY PRODUCT

product_summary = df.groupby(

  "Product Name"

)[["Sales", "Quantity"]].sum()

print("\n SALES AND QUANTITY BY PRODUCT ")

display(product_summary)

#BAR PLOT

plt.figure(figsize=(8, 5))

product_summary["Sales"].plot(kind="bar")
```

```
plt.title("Total Sales by Product")

plt.xlabel("Product Name")

plt.ylabel("Total Sales")

plt.xticks(rotation=0)

plt.tight_layout()

plt.show()

#LINE PLOT

df["Date"] = pd.to_datetime(df["Date"])

daily_sales = df.groupby("Date")["Sales"].sum()

plt.figure(figsize=(8, 5))

plt.plot(

   daily_sales.index,

   daily_sales.values,

   marker="o"

)

plt.title("Sales Over Time")

plt.xlabel("Date")

plt.ylabel("Sales")

plt.xticks(rotation=45)

plt.tight_layout()

plt.show()

#PIVOT TABLE

pivot_table = pd.pivot_table(

   df,

   values="Sales",

   index="Region",

   columns="Product Name",

   aggfunc="sum",

   fill_value=0

)

print("\n PIVOT TABLE ")
```

```
display(pivot_table)

#CORRELATION MATRIX

correlation = df[

   ["Sales", "Quantity"]

].corr()

print("\nCORRELATION MATRIX ")

display(correlation)

#HEATMAP

plt.figure(figsize=(6, 4))

sns.heatmap(

   correlation,

   annot=True,

   cmap="coolwarm",

   fmt=".2f"

)

plt.title("Correlation Heatmap")

plt.tight_layout()

plt.show()
```

Output:

```
Data file created successfully!
File name: sales_data.csv

 DATA FILE LOADED
```

	S.No	Date	Product Name	Sales	Quantity	Region
0	1	2026-08-01	Laptop	55000.0	5.0	Chennai
1	2	2026-08-02	Mobile	NaN	10.0	Coimbatore
2	3	2026-08-03	Tablet	20000.0	NaN	Madurai
3	4	2026-08-04	NaN	60000.0	6.0	Tiruchirappalli
4	5	2026-08-05	Mobile	35000.0	12.0	NaN

```
 SALES DATA
```

	S.No	Date	Product Name	Sales	Quantity	Region
0	1	2026-08-01	Laptop	55000.0	5.0	Chennai
1	2	2026-08-02	Mobile	NaN	10.0	Coimbatore
2	3	2026-08-03	Tablet	20000.0	NaN	Madurai
3	4	2026-08-04	NaN	60000.0	6.0	Tiruchirappalli
4	5	2026-08-05	Mobile	35000.0	12.0	NaN

```
MISSING VALUES
S.No            0
Date            0
Product Name    1
Sales           1
Quantity        1
Region          1
dtype: int64
```

DATA AFTER HANDLING MISSING VALUES

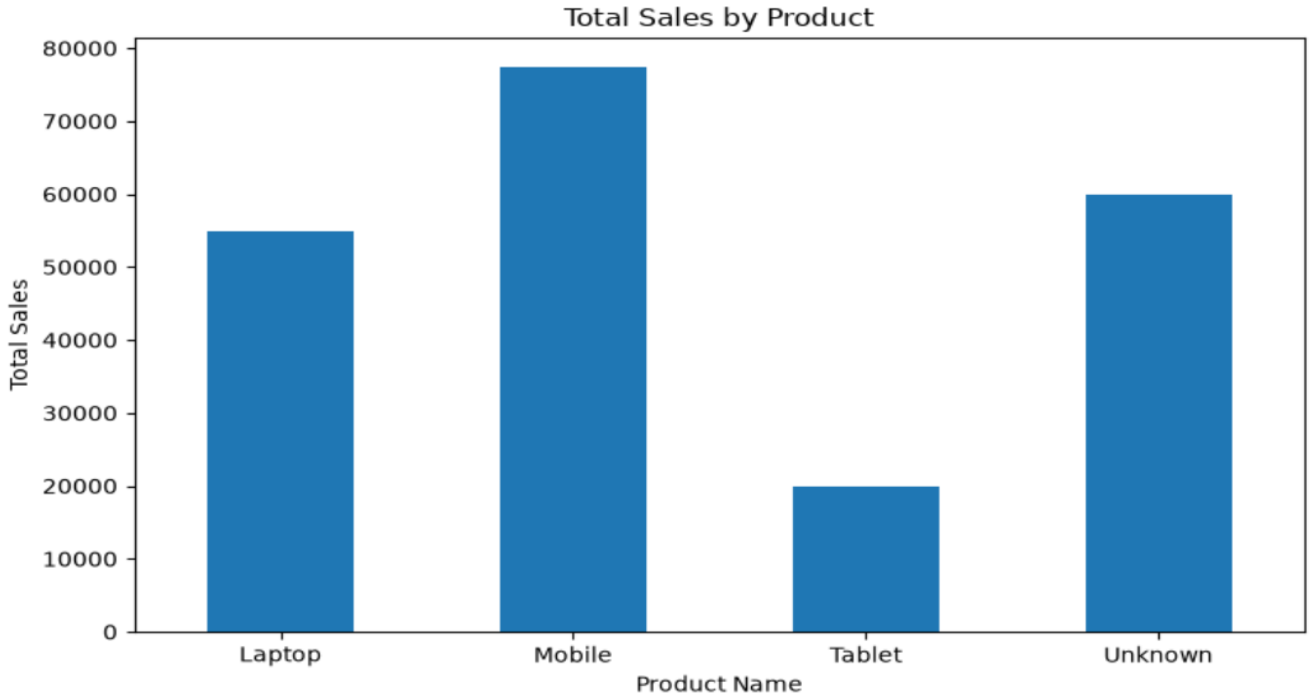
	S.No	Date	Product Name	Sales	Quantity	Region
0	1	2026-08-01	Laptop	55000.0	5.00	Chennai
1	2	2026-08-02	Mobile	42500.0	10.00	Coimbatore
2	3	2026-08-03	Tablet	20000.0	8.25	Madurai
3	4	2026-08-04	Unknown	60000.0	6.00	Tiruchirappalli
4	5	2026-08-05	Mobile	35000.0	12.00	Unknown

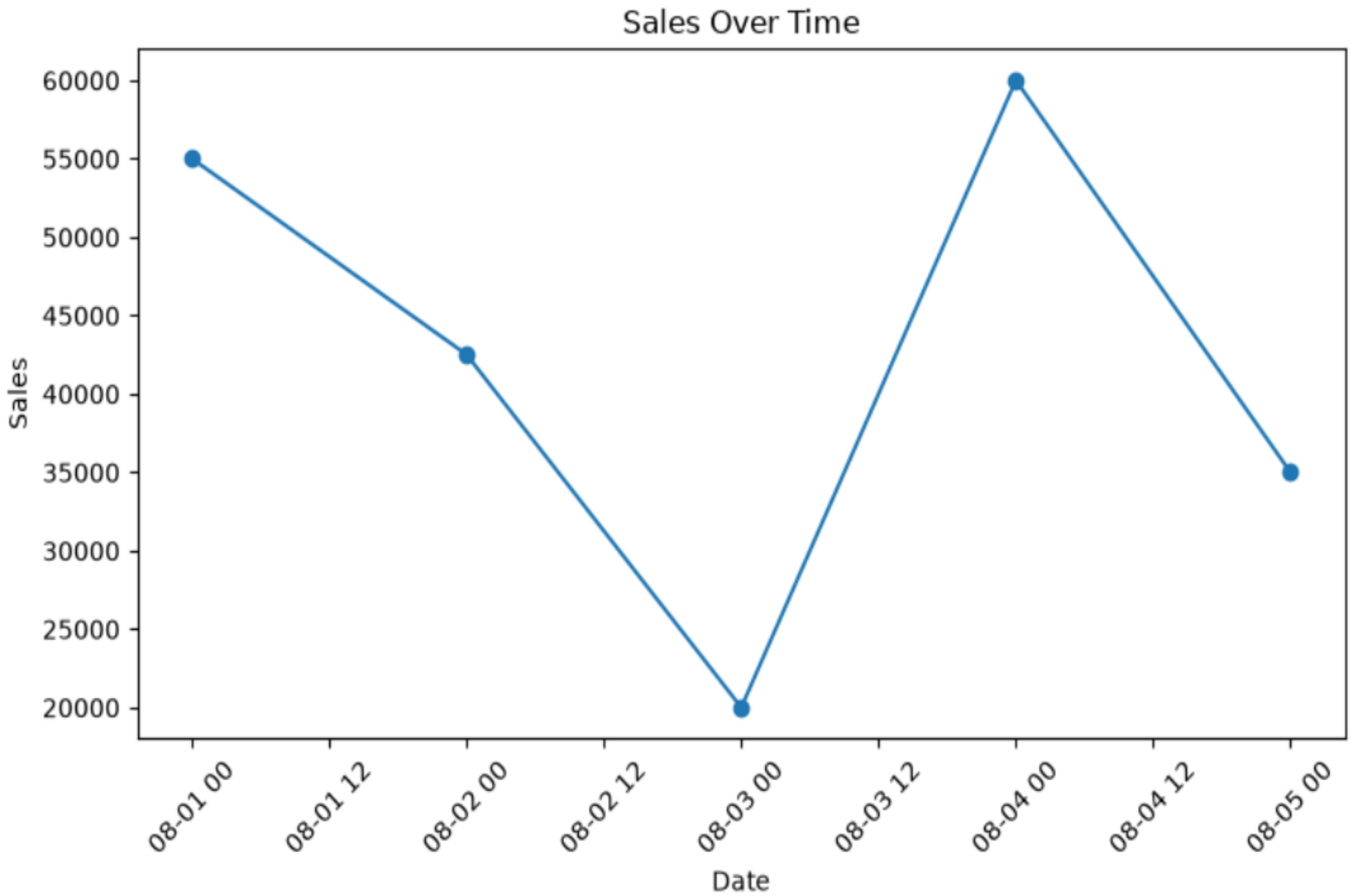
SUMMARY STATISTICS

	S.No	Sales	Quantity
count	5.000000	5.000000	5.000000
mean	3.000000	42500.000000	8.250000
std	1.581139	16007.810594	2.861381
min	1.000000	20000.000000	5.000000
25%	2.000000	35000.000000	6.000000
50%	3.000000	42500.000000	8.250000
75%	4.000000	55000.000000	10.000000
max	5.000000	60000.000000	12.000000

SALES AND QUANTITY BY PRODUCT

Product Name	Sales	Quantity
Laptop	55000.0	5.00
Mobile	77500.0	22.00
Tablet	20000.0	8.25
Unknown	60000.0	6.00


Total Sales by Product
80000
70000
60000
50000
40000
30000
20000
10000
0
Total Sales
Laptop
Mobile
Tablet
Unknown
Product Name


Sales Over Time
60000
55000
50000
45000
40000
35000
30000
25000
20000
Sales
08-01 00
08-01 12
08-02 00
08-02 12
08-03 00
08-03 12
08-04 00
08-04 12
08-05 00
Date

PIVOT TABLE

Product Name	Laptop	Mobile	Tablet	Unknown
Region				
Chennai	55000.0	0.0	0.0	0.0
Coimbatore	0.0	42500.0	0.0	0.0
Madurai	0.0	0.0	20000.0	0.0
Tiruchirappalli	0.0	0.0	0.0	60000.0
Unknown	0.0	35000.0	0.0	0.0

CORRELATION MATRIX

	Sales	Quantity
Sales	1.000000	-0.590145
Quantity	-0.590145	1.000000